Seminar infinitesimal and prismatic site

1 Overview of the infinitesimal site

Speaker: Adrian, 18/09/26

1.1 Topological spaces, presheaves and sheaves on them

Let X be a topological space. We have the category $\mathrm{Open}(X)$, where objects are open subsets $U \underset{i}{\hookrightarrow} X$ and for $U, V \subseteq X$, $\mathrm{Hom}(U,V) = \begin{cases} \{i\} & \text{if } U \subseteq V \\ \varnothing & \text{otherwise} \end{cases}$.

Definition 1. *1. A presheaf F on X is a contravariant functor $F : \mathrm{Open}(X) \to \mathrm{Sets}$.*

2. F is a sheaf on X if it is a presheaf and $\forall U \subseteq X$ open and $\{U_i\}_{i\in I}$ open cover of U, we have an exact sequence

$$F(U) \longrightarrow \prod_{i\in I} F(U_i) \rightrightarrows \prod_{(i,j)\in I^2} F(U_i \cap U_j)$$

Given an abelian sheaf on X, $X \in \mathrm{Ob}(\mathrm{Open}(X))$ is a final object, i.e. for every $U \in \mathrm{Ob}(\mathrm{Open}(X))$, there exists a unique $U \hookrightarrow X$. Then we can define global sections as $\Gamma(X,F)$ and $H^i(X,F) := R^i\Gamma(X,-)(F)$, with $R^i\Gamma(X,-)$ the ith right derived functor of $\Gamma(X,-)$.

Grothendieck: To define presheaves, sheaves and sheaf cohomology, we only need $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ satisfying certain axioms.

Definition 2. *A site is a pair $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ satisfying that:*

- *$\mathcal{C}$ a category with fibred products.*
- *$\mathrm{Cov}(\mathcal{C})$ a collection, for every $M \in \mathrm{Ob}(\mathcal{C})$ of morphisms $f_i : M_i \to M$ satisfying that*
  - *If $\{M_i \to M\}_{i\in I}$ is a cover of M and $\{M_{ij} \to M_i\}_{J_i}$ are covers of M_i, then $\{M_{ij} \to M\}_{i\in I, j\in J_i}$ is a cover of M.*
  - *$\mathrm{id} : M \to M$ is a cover of M*
  - *If $M' \to M$, and $\{M_i \to M\}_{i\in I}$ are a cover of M, then $\{M_i \times_M M' \to M'\}_{i\in I}$ is a cover of M'.*

The objects $M \in \mathrm{Ob}(\mathcal{C})$ play the role of open sets, the morphisms $M \to M'$ play the role of inclusions of open sets, and the coverings play the role of open covers.

A presheaf on $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ is a contravariant functor $\mathcal{C} \to \mathrm{Sets}$, and a presheaf is a sheaf if for every $M \in \mathrm{Ob}(\mathcal{C})$, and $\{M_i \to M\}_{i \in I} \in \mathrm{Cov}(M)$, we have that the following diagram is exact:

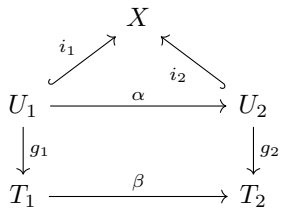
$$F(M) \longrightarrow \prod_{i \in I} F(M_i) \rightrightarrows \prod_{(i,j) \in I^2} F(M_i \cap M_j)$$

1.2 The infinitesimal site

Let K be a field of characteristic 0, and let X be a scheme (or a rigid space/adic space, in this case we must have $K/\mathbb{Q}_p$ finite extension). Suppose that X is smooth over K.

Definition 3. *The infinitesimal site of X consists of:*

- *As a category, we take $\mathcal{C}$ consisting of:*
  - *Objects: Pairs (U, T) where $i : U \hookrightarrow X$ is an open immersion (if we work with rigid spaces it should be an admissible open), and $g : U \to T$ is a nilpotent closed immersion, that is, it is a closed immersion of ideal $\mathcal{I} \subseteq \mathcal{O}_T$ and, $\forall t \in T$, there exists T' neighborhood of t such that $(\mathcal{I}|_{T'})^n = 0$ for some n. Observe that if T is quasi-compact, then this condition is equivalent to $\mathcal{I}^N = 0$ for some N. We think as T as a nilpotent thickening of U.*
  - *Morphisms: $(U_1, T_1) \to (U_2, T_2)$ consists of a pair (α, β) with $\alpha : U_1 \to U_2$ open immersion and $\beta : T_1 \to T_2$ is a morphism making the following diagram commutative:*



- *Coverings of an object (U, T) consist of a family $\{(\alpha_i, \beta_i) : (U_i, T_i) \to (U, T)\}$ such that $\{\alpha_i : U_i \to U\}$ is a open covering of U and $\{\beta_i : T_i \to T\}$ is a open covering of T.*

Thus defined, $\mathcal{C}$ is a category with fibred products and equalizers, and $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ is a site, denoted $(X/K)_{\mathrm{inf}}$ (this will be shown in 4).

Let X_{inf} denote the infinitesimal topos, that is, the category of sheaves of sets on $(X/K)_{\mathrm{inf}}$. This is the important category (according to Grothendieck), because different sites might give the same topos, so the topos is the right category to consider.

Theorem 1. *1. The data of a presheaf F on $(X/K)_{\mathrm{inf}}$ is equivalent to the data of, for every $(U, T) \in \mathrm{Ob}(\mathcal{C})$, a presheaf $F_{(U,T)}$ on T (for the usual topology, i.e. Zariski in the case of schemes), satisfying that, if we have a morphism $(\alpha, \beta) : (U_1, T_1) \to (U_2, T_2)$, then there is a morphism $\phi_{(\alpha,\beta)} : \beta^{-1} F_{(U_2,T_2)} \to F_{(U_1,T_1)}$ (and some additional property, we will see that in detail in 5).*

2. A presheaf is a sheaf on $(X/K)_{\mathrm{inf}}$ if and only if for every (U, T), $F_{(U,T)}$ is a sheaf.

Example 1. *$\mathcal{O}_{\text{inf}}$ the structure sheaf of $(X/K)_{\text{inf}}$, corresponds to the sheaf $\mathcal{O}_{\text{inf}(U,T)} = \mathcal{O}_T$.*

Now we want to discuss global sections. There is a problem, which is that, in general, the category $\mathcal{C}$ that we have defined doesn't have a final object. To solve this, instead of working with a final object in the site, we work with a final object in the topos.

Let $\mathbb{1}^{\text{pre}}(U,T) := \{*\}$ and $\mathbb{1}$ be the sheaffification of the presheaf $\mathbb{1}^{\text{pre}}$. Observe that, if F is a sheaf on $(X/K)_{\text{inf}}$, we have a canonical map $F \to \mathbb{1}^{\text{pre}}$ which is simply given by $F(U,T) \to \mathbb{1}^{\text{pre}}(U,T) = \{*\}$. Now by the formalism of sheaffification, we get a unique map $F \to \mathbb{1}^{\text{pre}} \to \mathbb{1}$. Then, if F is a sheaf on $(X/K)_{\text{inf}}$, we define global sections as $\Gamma((X/K_{\text{inf}}), F) := \text{Hom}_{X_{\text{inf}}}(\mathbb{1}, F)$. Then, we define cohomology as $H^i((X/K)_{\text{inf}}, -) := R^i\Gamma((X/K_{\text{inf}}, -))$. We also remark that there is an explicit description of $\Gamma(X/K_{\text{inf}}, F)$ using envelopes of closed immersions.

Why do all this? Let's shift to (apparently) annother topic.

Definition 4. *On X/K, a module with a connection is a pair $(\mathcal{F}, \nabla)$ where $\mathcal{F}$ is an $\mathcal{O}_X$-module and $\nabla : \mathcal{F} \to \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^1_{X/K}$ is a K-linear map satisfying $\forall U \subseteq X$ open, $a \in \mathcal{O}_X(U)$ and $b \in \mathcal{F}(U)$, $\nabla(ab) = a\nabla(b) + b \otimes da$.*

Given a connection, it induces a sequence

$$\mathcal{F} \underset{\nabla}{\to} \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^1_{X/K} \underset{\nabla^1}{\longrightarrow} \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^2_{X/K} \to \dots$$

via $\nabla^1(b \otimes \omega) = \nabla(b) \wedge \omega + b \otimes d\omega$, for $b \in \mathcal{F}(U)$ and $\omega \in \Omega^1_{X/K}$. Similarly we define ∇^2, ∇^3 and so on.

Definition 5. *We say that a connection ∇ is integrable if $\nabla^1 \circ \nabla = 0$.*

If ∇ is integrable, then the sequence above is a complex, denoted $\mathcal{F} \otimes_{\mathcal{O}_X} \Omega^\bullet_{X/K}$.

Definition 6. *Let X/K be smooth. Then, we define the de Rham cohomology as*

$$H^i_{\text{dR}}(X, (\mathcal{F}, \nabla)) := \mathbb{H}^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^\bullet_{X/K})$$

Theorem 2. *Let X/K be smooth, $\mathcal{F}$ be a locally free $\mathcal{O}_X$-module, and $(\mathcal{F}, \nabla)$ be an integrable connection. Then, we have:*

1. *There exists a unique sheaf F_{inf} on $(X/K)_{\text{inf}}$ associated to $(\mathcal{F}, \nabla)$.*
2. *$H^i_{\text{dR}}(X, (\mathcal{F}, \nabla)) \cong H^i((X/K)_{\text{inf}}, F_{\text{inf}})$.*

This works as follows: Given $(\mathcal{F}, \nabla)$ on X, and $(U,T) \in \text{Ob}(\mathcal{C})$, $g : U \to T$ and $T' \geq T$, we have a lift by definition of smoothness $\phi_{T'} : T' \to U$. Then we can define $F_{T'} = \phi^*_{T'}\mathcal{F}$. In general these sheaves don't glue, but the connection allows to glue them and form a sheaf $(F_{\text{inf}})_{(U,T)}$.

Remark 1. *The de Rham cohomology comes with a filtration. The construction of de Rham cohomology using the infinitesimal site does not give the filtration if we work over K. However it can be recovered working in the more general setting of B^+_{dR}-coefficients (see [7]).*

Example 2. *If X is smooth and proper, the diagonal embedding $\Delta : X \to X \times_K X$ is a closed immersion, and it factors through the nth infinitesimal neighborhood of the diagonal $\mathcal{P}^n_{X/K}$, for every*

n. We have that $(X, \mathcal{P}^n_{X/K}) \in \mathrm{Ob}(\mathcal{C})$. In the affine case, $X = \operatorname{Spec} B$, $X \to X \times_K X$ corresponds to the multiplication map $B \otimes_K B \to B$ given by $b \otimes b' \mapsto bb'$. Let I be the kernel of this map. Then $\mathcal{P}^n_{X/K} = \operatorname{Spec}(B \otimes_K B/I^{n+1})$, and the first order thickening corresponds to the sequence

$$0 \to I/I^2 \to B \otimes_K B/I^2 \to B \to 0$$

This should give us an idea of why we can compute de Rham cohomology using that site.

2 Differential operators

Speaker: Oriol, 25/09/26

Definition 7. *Let X be a topological space, and $\mathcal{O}_1 \to \mathcal{O}_2$ sheaves of rings on X. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_2$-modules and $k \geq 0$. A differential operator of order $\leq k$, $D \in \mathrm{Diff}^k_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$ is a $\mathcal{O}_1$-linear map $D : \mathcal{F} \to \mathcal{G}$ that satisfies the following property. For open $U \subseteq X$, and $a \in \mathcal{O}_2(U)$, the map*

$$s \mapsto D(as) - aD(s)$$

is a differential operator of order $\leq k-1$. We define a differential operator of order 0 to be a $\mathcal{O}_2$-linear map.

Observation 1.

1. *$\mathrm{Diff}^k_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$ form a $\mathcal{O}_2$-module. This is clear doing induction on k and observing that for $k = 0$ we have $\mathrm{Diff}^0_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G}) = \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{F}, \mathcal{G})$ which is a $\mathcal{O}_2$-module.*
2. *If $n \geq m$, we have $\mathrm{Diff}^n_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G}) \supseteq \mathrm{Diff}^m_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$. Also clear from induction and the fact that the zero map is linear (so a linear map is a differential operator of order ≤ 1).*

Now let's see what this recursive definition means in concrete terms, doing the cases $k = 1$ and $k = 2$.

k=1 Let $s \in \mathcal{F}(U)$ and $a, b \in \mathcal{O}_2(U)$. Then a differential operator of order 1 satisfies that $s \mapsto D(as) - aD(s)$ is linear, so we have $D(abs) - aD(bs) = b(D(as) - aD(s))$, which rearanging terms gives

$$D(abs) = aD(bs) + bD(as) - abD(s)$$

k=2 Let $s \in \mathcal{F}(U)$ and $a, b, c \in \mathcal{O}_2(U)$. Then a differential operator of order 2 satisfies that $D_a := D(as) - aD(s)$ is a differential operator of order 1, so we have, using what we have just obtained above, that $D_a(bcs) = bD_a(cs) + cD_a(bs) - bcD_a(s)$ and substituting the expression of D_a we get $D(abcs) - aD(bcs) = b(D(acs - aD(cs))) + c(D(abs) - aD(bs)) - bc(D(as) - aD(s))$ and rearanging terms:

$$D(abcs) = aD(bcs) + bD(acs) + cD(abs) - abD(cs) - bcD(as) - acD(bs) + abcD(s)$$

Example 3. *If we let $\mathcal{F} = \mathcal{O}_2$, then we can let $s = 1$ in the formula for $k = 1$ obtained above, and we end up with*

$$D(ab) = aD(b) + bD(a) - abD(1)$$

Now define a new map $d : \mathcal{O}_2 \to \mathcal{G}$ by $d(a) = D(a) - aD(1)$, and we have that $d(ab) = D(ab) - abD(1) = aD(b) + bD(a) - abD(1) - abD(1) = a(D(b) - bD(1)) + b(D(a) - aD(1))$. So $d(ab) = ad(b) + bd(a)$ and d is a derivation. Now $D(1) \in \mathcal{G}(U)$, and D is completely determined by d and $D(1)$, so we have

$$\mathrm{Diff}^1(\mathcal{O}_2, \mathcal{G}) \cong \mathrm{Der}_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{G}) \oplus \mathcal{G}$$

From now on, we specialize to the situation of schemes: Let $f : X \to S$ be a morphism of schemes that we will assume to be separated. Then we have two sheaves of rings on X, $f^{-1}(\mathcal{O}_S) \to \mathcal{O}_X$. We want to give an equivalent but more geometric notion of what is a differential operator. The rest of this section follows [3].

Observe that given a $f^{-1}(\mathcal{O}_X)$-linear map $D : \mathcal{F} \to \mathcal{G}$ we can make it $\mathcal{O}_X$-linear it by setting $\overline{D} : \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \to \mathcal{G}$. Now via the isomorphism $\mathcal{F} \cong \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F}$, we have $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \cong \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F}$, so $\overline{D}$ can be seen as the map $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F} \to \mathcal{G}$ that is given on sections by $\overline{D}(a \otimes b \otimes s) = aD(bs)$.

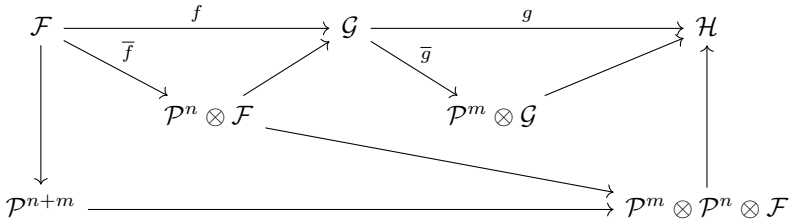
Notation 1. *From now on denote $\mathcal{P} = \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X$ and $\mathcal{I} = \ker(\mathcal{P} \xrightarrow{m} \mathcal{O}_X)$ where m is the multiplication map $a \otimes b \mapsto ab$. In other words, $\mathcal{I}$ is the ideal defining the diagonal immersion $X \to X \times_S X$. We will also denote $\mathcal{P}^n = \mathcal{P}/\mathcal{I}^{n+1}$.*

Proposition 1. *$D \in \mathrm{Diff}^k_{X/K}(\mathcal{F}, \mathcal{G})$ if and only if $\overline{D}$ factors through $\mathcal{P} \otimes \mathcal{F} \to \mathcal{P}^k \otimes \mathcal{F} \to \mathcal{G}$.*

Proof. We proceed by induction on k. For $k = 0$, observe that D is $\mathcal{O}_X$-linear if and only if $sD(x) = D(ax)$ for every $a \in \mathcal{O}_X(U)$, which in terms of $\overline{D}$ is equivalent to $\overline{D}(a \otimes 1 \otimes x) = \overline{D}(1 \otimes a \otimes x)$, or equivalently, $\overline{D}((a \otimes 1 - 1 \otimes a) \otimes x) = 0$. Then as $\mathcal{I}$ is generated by elements of the form $a \otimes 1 - 1 \otimes a$ this is equivalent to $\overline{D}$ factoring through $\mathcal{P}^1$. Now assume that the statement is true for $k-1$. Then $D \in \mathrm{Diff}^k_{X/S}(\mathcal{F}, \mathcal{G})$ if and only if $\forall a$, $D_a \in \mathrm{Diff}^{k-1}(\mathcal{F}, \mathcal{G})$ with $D_a(x) = D(ax) - aD(x)$. But then by induction hypothesis this means that $\overline{D_a}$ factors through $\mathcal{P}^{k-1}$, that is $\overline{D_a}(t \otimes x) = 0$, for every $t \in I^k$. But this is equivalent to $\overline{D}((1 \otimes a)t \otimes x - (a \otimes 1)t \otimes x) = 0$ and we're done as every element of $\mathcal{I}^{k+1}$ can be written as $(a \otimes 1 - 1 \otimes a)t$ for $t \in \mathcal{I}^k$. □

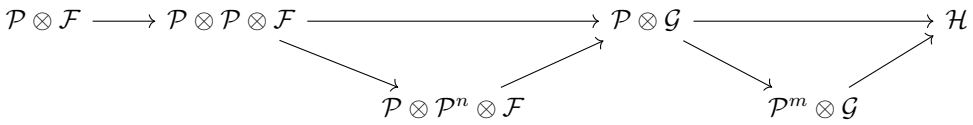
Remark 2.
1. *Proposition 1 tells us that a differential operator can be regarded as a $f^{-1}(\mathcal{O}_S)$-linear map that is "almost $\mathcal{O}_X$-linear".*
2. *We have* $\mathrm{Diff}^k_{X/S}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}^k \otimes \mathcal{F}, \mathcal{G})$

Proposition 2. *Let $f \in \mathrm{Diff}^n_{X/K}(\mathcal{F}, \mathcal{G})$ and $g \in \mathrm{Diff}^m_{X/K}(\mathcal{G}, \mathcal{H})$. Then $g \circ f \in \mathrm{Diff}^{n+m}_{X/K}(\mathcal{F}, \mathcal{H})$ and we have a commutative diagram*



Proof. Observe that the only thing that we need to prove is that the diagram commutes, as then $\overline{g \circ f}$ does indeed factor through $\mathcal{P}^{n+m} \otimes \mathcal{F}$ and so and therefore we have $g \circ f \in \mathrm{Diff}^{n+m}_{X/S}(\mathcal{F}, \mathcal{G})$. Then observe that we have $\overline{g \circ f} = \overline{g} \circ (\mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f)$ as both map $a \otimes b \otimes s \mapsto a(g \circ f)(bs)$. Now we express $(\mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f)$ in terms of $\overline{f}$: For that, consider the map $\delta : \mathcal{P} \to \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{P}$ given on sections by $a \otimes b \mapsto a \otimes 1 \otimes 1 \otimes b$. This is a ring homomorphism. Then consider the composition $\overline{\delta}(f) : \mathcal{P} \otimes \mathcal{F} \xrightarrow[\delta \otimes \mathrm{id}_F]{} \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{F} \xrightarrow[\mathrm{id}_{\mathcal{P}} \otimes \overline{f}]{} \mathcal{G}$. Via the isomorphism $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \cong \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{F}$ we

have $\overline{\delta}(f) = \mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f$, as we can check on sections: $\overline{\delta}(f)(a \otimes b \otimes s) = (\mathrm{id}_{\mathcal{P}} \otimes \overline{f})(a \otimes 1 \otimes 1 \otimes b \otimes s) = (a \otimes 1) \otimes f(bs)$. Therefore it follows that $\overline{g} \circ \overline{\delta}(f) = \overline{g \circ f}$. This gives us a diagram



which tells us that $\overline{g \circ f}$ factors through $\mathcal{P}^m \otimes \mathcal{P}^n \otimes \mathcal{F}$. Therefore, the only thing we need to check is that $\delta_{m,n} : \mathcal{P} \underset{\delta}{\to} \mathcal{P} \otimes \mathcal{P} \to \mathcal{P}^m \otimes \mathcal{P}^n$ factors through $\mathcal{P}^{m+n}$. For that, observe that $\mathcal{I}$ is generated by elements $\eta = 1 \otimes x - x \otimes 1$, and $\delta(\eta) = 1 \otimes 1 \otimes 1 \otimes x - x \otimes 1 \otimes 1 \otimes 1$. Now the tensor product in the middle is with respect to $\mathcal{O}_X$, so $1 \otimes x \otimes 1 \otimes 1 = 1 \otimes 1 \otimes x \otimes 1$, and so we can write $\delta(\eta) = \mathbb{1} \otimes \eta + \eta \otimes \mathbb{1}$, where $\mathbb{1} = 1 \otimes_{f^{-1}(\mathcal{O}_S)} 1$. Now as δ is a ring homomorphism, every generator of $\mathcal{I}^{n+m+1}$ will be sent to $\prod_{i=1}^{n+m+1}(\mathbb{1} \otimes \eta_i + \eta_i \otimes \mathbb{1})$ and every term when we expand the product will have either $m+1$ η's in the first $\mathcal{P}$-coordinate or $n+1$ in the second, so it will vanish modulo $\mathcal{I}^{m+1} \otimes \mathcal{I}^{n+1}$. □

Proposition 3. *Let X/S be smooth and $x_1, \ldots, x_n$ be local coordinates, that is, we have an étale map $g : X \to \mathbb{A}^n_S$ defined by the sections $x_1, \ldots, x_n \in \mathcal{O}_X(X)$. Denote $\eta_i = 1 \otimes x_i - x_i \otimes 1$. Then*

1. *$\mathcal{P}^n$ is a free $\mathcal{O}_X$-module with basis the images of $\{\eta_1^{\alpha_1} \cdots \eta_n^{\alpha_n}$ such that $\sum \alpha_i \leq n\}$.*
2. *Denote $q = (q_1, \ldots, q_n)$ and let $\overline{D}_q$ be the dual basis of η_q in $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P} \otimes \mathcal{F}, \mathcal{G})$. Let D_q be the corresponding differential operators. Then $\{D_q$ for $|q| \leq m\}$ is a basis of $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X, \mathcal{O}_X)$ and the composition is given by*

$$D_q \circ D_{q'} = \binom{q+q'}{q} D_{q+q'}$$

where the facorial of a multiindex is defined by $q! = \prod q_i!$.

Proof. 1. It follows from results on regular sequences. There are proofs of some of the statements about regular sequences used in [5]. Recall that for a ring A, and elements $a_i \in A$, $(a_1, \ldots, a_n)$ is a regular sequence if, for every i, a_i is not a zero divisor in $A/(a_1, \ldots, a_{i-1})$. We say that a morphism of schemes $f : X \to Y$ is a regular immersion if $\forall x \in X$, $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ has kernel generated by a regular sequence. A surjective morphism of local regular rings has kernel generated by a regular sequence. Then, in our situation, the diagonal embedding $X \to X \times_S X$ is closed so the local maps are surjective, and $f : X \to S$ is smooth, so the local rings are regular and therefore $X \to X \times_S X$ is a regular immersion.

Now we have the following lemma:

Lemma 1. *If $\mathcal{I}$ is an ideal of $\mathcal{O}_X$ that is locally generated by a regular sequence, it satisfies that $\mathcal{I}^j/\mathcal{I}^{j+1} \cong \mathrm{Sym}^j(\mathcal{I}/\mathcal{I}^2)$.*

Observe that the étale map g gives an isomorphism $g^*\Omega_{\mathbb{A}^n_S/S} \cong \Omega^1_{X/S}$, and therefore $\Omega^1_{X/S}$ is free generated by the images of η_i. Then by the lemma above $\mathcal{I}^j/\mathcal{I}^{j+1}$ is also free generated

by the images of $\eta_1^{\alpha_1}\cdots\eta_n^{\alpha_n}$ with $\sum\alpha_i = j$. Then we can do induction on the sequences (which split because $\mathcal{P}^{j-1}$ is free by induction hypothesis, and therefore projective)

$$0 \to \mathcal{I}^j/\mathcal{I}^{j+1} \to \mathcal{P}^j \to \mathcal{P}^{j-1} \to 0$$

and we conclude that $\mathcal{P}^j$ is free for every j and the elements $\{\eta_1^{\alpha_1}\cdots\eta_n^{\alpha_n}$ such that $\sum\alpha_i \leq n\}$ form a basis.

2. It is clear that the D_q's form a basis of $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X,\mathcal{O}_X)$, so we just need to check the statement about the composition. For that, it's enough to check how $\overline{D_q\circ D_{q'}} = \overline{D_q}\circ(\mathrm{id}_{\mathcal{P}}\otimes\overline{D_{q'}})\circ\delta$ act on η^p (recall here p is a multiindex). Recall from the proof of Proposition 2 that δ is a ring homomorphism, and so we have

$$\delta(\eta^p) = (\eta\otimes\mathbb{1}+\mathbb{1}\otimes\eta)^p = \sum_{i+j=p}\binom{p}{i}(\eta\otimes\mathbb{1})^i(\mathbb{1}\otimes\eta)^j$$

Now applying $\overline{D_{q'}}$ only the terms with $j=q'$ survive because of the dual basis definition, and the same with $\overline{D_q}$, only the terms with $i=q$ survive, and we end up with the formula of the statement.

□

Remark 3. *(about Proposition 3)*

1. *It seems that this still works even if $X\to S$ is not separated. Then the diagonal map is still a locally closed immersion. However the ideal $\mathcal{I}$ may not be defined in all X, only locally, but $\mathcal{I}^j/\mathcal{I}^{j+1}$ can be defined in all X because they're naturally $\mathcal{O}_X$-modules.*
2. *The result implies in particular that the D_q's commute.*
3. *If S is a scheme over $\mathbb{Q}$, then we are able to invert the factorials, and so $\mathrm{Diff}^1_{X/S}(\mathcal{O}_X,\mathcal{O}_X)$ generate $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X,\mathcal{O}_X)0$ for every m.*

Example 4. *Let $X=\mathbb{A}^2_K$, for K a field of characteristic 0. Then, every differential operator $D\in\mathrm{Diff}^k_{\mathbb{A}^2_K/K}(\mathcal{O}_X,\mathcal{O}_X)$ is of the form*

$$D = \sum_{m+n\leq k}\frac{\partial^{m+n}}{\partial x^n\partial y^m}$$

To see that, given the result of Proposition 3, it is enough to see what are D_0, D_x and D_y, the differential operators corresponding to the dual basis of $1, x\otimes 1-1\otimes x, y\otimes 1-1\otimes x$. For that it is enough to see what they do on monomials x^ny^m.

- *$D_0(1)=1$, and $0=\overline{D_0}(1\otimes x - x\otimes 1) = D_0(x)-xD_0(1)$, so $D_0(x)=x$ and similarly $D_0(y)=y$. Now by induction it's easy to see that $D_0(x^my^n)=x^ny^m$. Assume for instance $n\geq 1$, and that the result holds for $n'+m'<n+m$. Then as it is a order 1 differential operator $0=\overline{D_0}(x^{n-1}y^m\otimes 1-1\otimes x^{n-1}y^m)(x\otimes 1-1\otimes x)) = D(x^ny^m)-xD_0(x^{n-1}y^m)-x^{n-1}y^mD_0(x)+x^ny^mD_0(1)$ and so $D_0(x^ny^m)=x^ny^m$.*

- $D_x(1) = 0$ *and* $1 = \overline{D_x}(1 \otimes x - x \otimes 1) = xD_x(1) - D_x(x)$ *so* $D_x(x) = 1$ *and the same shows* $D_x(y) = 0$. *Now by induction again we can show* $D_x(x^n y^m) = nx^{n-1}y^m$ *for* $n \geq 1$ *and* $D_x(y^m) = 0$. *Again by being a order 1 differential operator, we get* $0 = \overline{D_x}(x^{n-1}y^m \otimes 1 - 1 \otimes x^{n-1}y^m)(x \otimes 1 - 1 \otimes x)) = D(x^n y^m) - xD_x(x^{n-1}y^m) - x^{n-1}y^m D_x(x) + x^n y^m D_x(1)$ *and by induction hypothesis we have* $D(x^n y^m) = xD_x(x^{n-1}y^m) + x^{n-1}y^m D_x(x) = x(n-1)x^{n-2}y^m + x^{n-1}y^m = nx^{n-1}y^m$. *Therefore* $D_x = \frac{\partial}{\partial x}$

The symmetric reasoning shows $D_y = \frac{\partial}{\partial y}$. *In conclusion, every differential operator or order* ≤ 1 *can be written as*

$$D = P(x,y)\frac{\partial}{\partial x} + Q(x,y)\frac{\partial}{\partial y} + R(x,y)$$

and as already said, the more general statement follows because D_x *and* D_y *generate* $D_{(n,m)}$ *for higher order differential operators.*

2.1 Connections as differential operators

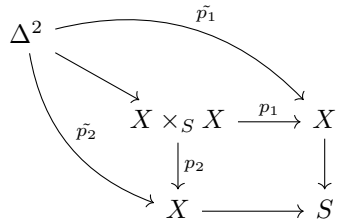
Definition 8. *Let* $X \to S$ *be a morphism of schemes, and* $\mathcal{F}$ *a* $\mathcal{O}_X$*-module. Then, a connection on* $\mathcal{F}$ *is a map*

$$\nabla : \mathcal{F} \to \mathcal{F} \otimes \Omega^1_{X/S}$$

satisfying, on sections, for every $a \in \mathcal{O}_X(U)$ *and* $x \in \mathcal{F}(U)$, $\nabla(ax) = a\nabla(x) + x \otimes da$.

Observation 2. *Observe that a connection is in particular a order 1 differential operator. Indeed,* $s \mapsto \nabla(as) - a\nabla(s) = s \otimes da$ *which is clearly* $\mathcal{O}_X$*-linear:* $bs \mapsto bs \otimes da = b(s \otimes da)$.

We have a more geometric equivalent definition of connection: Denote Δ^2 the closed subscheme of $X \times_S X$ defined by the ideal $\mathcal{I}^2$, that is a first order thickening of X. Then we have maps



Proposition 4. *A connection* $\nabla : \mathcal{F} \to \mathcal{F} \otimes \Omega^1_{X/S}$ *is equivalent to a* $\mathcal{P}^1_{X/S}$*-linear isomorphism* $\epsilon : \tilde{p_2}^*\mathcal{F} \to \tilde{p_1}^*\mathcal{F}$.

Proof. Given ϵ, define $\theta : \mathcal{F} \to \mathcal{F} \otimes \mathcal{P}^1$ by $\theta = \epsilon \circ p_2^{\#}$. Then we have that θ is linear for the right $\mathcal{O}_X$-structure on $\mathcal{P}$, that is, we have $\theta(ax) = (1 \otimes a)\theta(x)$. Now consider ∇ defined by $\nabla(x) = \theta(x) - x \otimes \mathbb{1}$. Now ϵ reduces to id $\mod \mathcal{I}$, which tells us that ∇ has image in $\mathcal{I}/\mathcal{I}^2 = \Omega^1_{X/S} \subseteq \mathcal{P}^1$, so the target is well defined. Now we just need to check that it satisfies the correct condition on sections. But indeed, $\nabla(ax) = (1 \otimes a)\theta(x) - ax \otimes \mathbb{1} = (1 \otimes a)\theta(x) + (1 \otimes a)(x \otimes \mathbb{1}) - (1 \otimes a)(x \otimes \mathbb{1}) - ax \otimes \mathbb{1} = (1 \otimes a)\nabla(x) + da \otimes x$.

Reciprocally, given a connection ∇, let $\theta(x) = \nabla(x) + x \otimes \mathbb{1}$. The same computation shows that θ is linear for the right $\mathcal{O}_X$ structure on $\mathcal{P}^1$. Then tensoring with θ we get a $\mathcal{P}^1$-linear map $\epsilon : \tilde{p_2}^*\mathcal{F} = \mathcal{P}^1 \otimes \mathcal{F} \to \mathcal{F} \otimes \mathcal{P}^1 = \tilde{p_1}^*\mathcal{F}$. and it clearly gives the identity modulo $\mathcal{I}$, by construction.

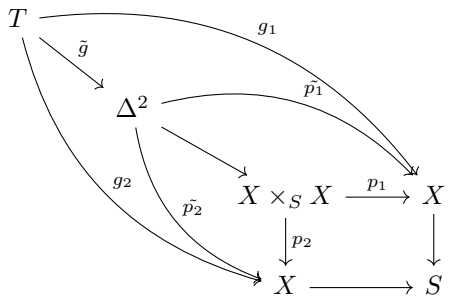
Now consider the involution $\tau : \mathcal{P}^1 \to \mathcal{P}^1$ given by $a \otimes b \mapsto b \otimes a$. This induces, tensoring with $\mathcal{F}$, a τ-linear map $\sigma : p_1^*\mathcal{F} \to p_2^*\mathcal{F}$, and gives a τ-linear map $\sigma \circ \epsilon : p_2^*\mathcal{F} \to p_2^*\mathcal{F}$. Now $(\sigma \circ \epsilon)^2$ is $\mathcal{P}^1$-linear, and reduces to the identity $\mod \mathcal{I}$. Then by Lemma 2 $(\sigma \circ \epsilon)^2$ is an isomorphism, so $\sigma \circ \epsilon$ is bijective and so is ϵ. This shows that ϵ is actually an isomorphism. □

Lemma 2. *Let $\varphi : M \to M$ be a A-linear map that reduces to* id *modulo an ideal $I \subseteq A$ such that $I^2 = 0$. Then φ is an isomorphism.*

Proof. Simply let $h(x) = \varphi(x) - x$. Then $h : M \to IM$. Consider then $\psi = x - h(x)$. We have $\psi \circ \varphi(x) = \psi(x + h(x)) = x + h(x) - h(x + h(x)) = x$, and similarly $\varphi \circ \psi = \mathrm{id}$, so φ is an isomorphism. □

Remark 4. *Proposition 4 tells us that a connection allows us to descend a sheaf on X to the first order thickening of X, Δ^2. That is, via the retractions $\tilde{p_i}^*$, we have in general two ways of pulling back the section $\mathcal{F}$ from X to Δ^2. But if $\mathcal{F}$ has a connection we have a way of identifying these pullbacks, and so a unique way to descend $\mathcal{F}$ to Δ^2 (up to isomorphism).*

More generally, if we have any other first order thickening of X, $X \to T$, and there exists a retraction $g_1 : T \to X$, then we have a unique (up to iso) way of descending $\mathcal{F}$ to T if $\mathcal{F}$ has a connection. Indeed, if there is any other retraction $g_2 : T \to X$, then $g = (g_1, g_2) : T \to X \times_S X$ will factor through Δ^2. Then we can pullback the isomorphism ϵ from Δ^2 to T along $\tilde{g}$ and we have a way of identifying the pullbacks $g_1^*\mathcal{F}$ and $g_2^*\mathcal{F}$.



Example 5. *What is the corresponding map ϵ for the connection given by the exterior derivative $d : \mathcal{O}_X \to \Omega^1_{X/S}$? Well, $d(x) = 1 \otimes x - x \otimes 1$ and so $\theta(x)$ is just the image of $1 \otimes x$ in $\mathcal{P}^1$. Now ϵ is the $\mathcal{P}^1$-linearization of θ, so $\epsilon(a \otimes b) = (a \otimes 1)\epsilon(1 \otimes b) = (a \otimes 1)(1 \otimes b) = a \otimes b$, so ϵ is just the identity map.*

3 Connections

Speaker: Paola, 02/10/26

3.1 De Rham Cohomology

Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and $\mathcal{A}$ with enough injectives, $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Let $\mathcal{D}(\mathcal{A})$ be the category of bounded from below complexes: Objects in $D(\mathcal{A})$ are complexes

$$M^\bullet : \cdots \to M^i \to M^{i+1} \to \dots$$

such that there exists some n_0 with $M^i = 0\ \forall i < n_0$. Then F induces a functor $D(F) : \mathcal{D}(\mathcal{A}) \to \mathcal{D}(\mathcal{B})$, and we call hyperderived functors of F the derived functors of $D(F)$. More precisely:

Definition 9. *If $A^\bullet \in \mathcal{D}(\mathcal{A})$, an injective resolucion is a map $0 \to A^\bullet \xrightarrow{\phi} I^\bullet$ such that*

a ϕ is injective.

b I^n is injective in $\mathcal{A}$ for all n

c ϕ is a quasi-isomorphism, that is, $H^n(\phi) : H^n(A^\bullet) \to H^n(I^\bullet)$ is an isomorphism.

We define the hyperderived functors of F as $\mathbb{R}^i F(M^\bullet) := H^i(F(I^\bullet))$.

Remark 5. *The definition of injective resolution in the category $D(\mathcal{A})$ is made such that $\mathbb{R}^n F(M^\bullet)$ is independent, up to canonical isomorphism, of the injective resolution.*

In particular, we are interested in the following situation:

Definition 10. *Let X be a scheme, and let $\mathcal{F}^\bullet : \mathcal{F}^0 \to \mathcal{F}^1 \to \dots$ be a complex of abelian sheaves on X. Then the ith hypercohomology of $\mathcal{F}^\bullet$ is $\mathbb{H}^i(X, \mathcal{F}) := \mathbb{R}^i\Gamma(X, -)(\mathcal{F})$. In particular, for the particular case of the de Rham complex $\Omega_X^\bullet : \mathcal{O}_X \to \Omega_X^1 \to \dots$, we have the de Rham cohomology $H^i_{\mathrm{dR}}(X) = \mathbb{H}^i(X, \Omega_X^\bullet)$.*

Now we'll see a recipie to calculate the de Rham cohomology of a separated, noetherian scheme X, and $\mathcal{F}^\bullet$ in $\mathrm{QCoh}(X)$. We first need a lemma.

Lemma 3. *Let $\varphi : M^\bullet \to N^\bullet$ be a quasi-isomorphism of bounded complexes such that N^k is F-acyclic, i.e. $R^iF(N^k)$, for all $i \geq 1$ and for all k. Then $\mathbb{R}^n F(M^\bullet) \cong H^n F(N^\bullet)$.*

Remark 6. *The content of the above lemma is that we can compute hyperderived functors using acyclic resolutions.*

Proof. Let $M^\bullet \to N^\bullet$ be a quasi-isomorphism. Then we have that $\mathbb{R}^i F(M^\bullet) = \mathbb{R}^i F(N^\bullet)$ (I think this is not immediate, but we showed it in p-adic Hodge theory). So, it suffices to prove that for $M^\bullet$ F-acyclic we have $\mathbb{R}^i F(M^\bullet) \cong H^i(F(M^\bullet))$. Now let $M^\bullet \to I^\bullet$ be an injective resolution and consider the short exact sequence $0 \to M^\bullet \to I^\bullet \to Q^\bullet \to 0$, that is, for every k, we have $0 \to M^k \to I^k \to Q^k \to 0$ exact. Now we take R^iF on this sequence, and we have $\forall i \geq 1$, the exact sequence

$$0 \to F(M^k) \to F(I^k) = 0 \to F(Q^k) \to R^iF(M^k) = 0$$

And therefore this shows that Q^k is F-acyclic for every k. Now taking cohomology we get an exact sequence

$$H^i(N^\bullet) \cong H^i(I^\bullet) \to H^i(Q^\bullet) \to H^{i+1}(M^\bullet) \cong H^{i+1}(I^\bullet)$$

and therefore this shows that $Q^\bullet$ is an exact complex. Now being exact and F-acyclic implies that $F(Q^\bullet)$ is exact as well, and therefore taking the long exact sequence associated in cohomology to the short exact sequence $0 \to F(M^\bullet) \to F(I^\bullet) \to F(Q^\bullet) \to 0$ we get $H^n(F(Q^\bullet)) = 0$ and therefore $H^n(F(M^\bullet)) = H^n(F(I^\bullet)) = \mathbb{R}^n(F(I^\bullet))$. □

Now let's go back to the goal of calculating the de Rham cohomology of a separated noetherian scheme X. Let $\mathcal{F}^\bullet$ be a complex with $\mathcal{F}^i \in \mathrm{QCoh}(X)$. Let $\mathfrak{U} = \{U_i\}_{i\in I}$ be a finite affine cover of X. Recall the Čech complex associated to $\mathcal{F} = \mathcal{F}^i$ for every fixed i.

$$\cdots \to \check{C}^p(\mathfrak{U},\mathcal{F}) = \prod_{i_0<\cdots<i_p} \mathcal{F}(U_{i_0}\cap\cdots\cap U_{i_p}) \xrightarrow[\delta^p]{} checkC^{p+1}(\mathfrak{U},\mathcal{F}) = \prod_{i_0<\cdots<i_{p+1}} \mathcal{F}(U_{i_0}\cap\cdots\cap U_{i_{p+1}}) \to \dots$$

with $\delta_p(a)_{i_0,\dots,i_{p+1}} = \sum_{k=0}^{p+1}(-1)^k a_{i_0,\dots,\hat{i_k},\dots i_{p+1}}|_{U_0\cap\cdots\cap U_{p+1}}$. Now let's denote for simplicity $U_{i_0,\dots,i_p} = U_{i_0}\cap\cdots\cap U_{i_p} \hookrightarrow [j_{i_0,\dots,i_p}]X$. Observe that we can sheaffify the Čech complex, defining $\check{\mathcal{C}}^p(\mathfrak{U},\mathcal{F}) = \prod_{i_0<\cdots<i_p} j_{i_0,\dots,i_p *}j^*_{i_0,\dots,i_p}\mathcal{F}$. This sheaf satisfies $\Gamma(X,\check{\mathcal{C}}^p(\mathfrak{U},\mathcal{F})) = \check{C}^p(\mathfrak{U},\mathcal{F})$. Therefore, using that the scheme is separated and that $\mathfrak{U}$ is an open affine cover, we have that

$$H^n(X,\check{\mathcal{C}}^\bullet(\mathfrak{U},\mathcal{F})) = \bigoplus H^n(U_{i_0,\dots,i_p},\mathcal{F}|_{i_0,\dots,i_p}) = 0 \quad \text{for } n>0$$

which means that $\check{\mathcal{C}}^p(\mathfrak{U},\mathcal{F})$ is Γ-acyclic. Then,

$$0 \to \mathcal{F} \to \check{\mathcal{C}}^p(\mathfrak{U},\mathcal{F})$$

is exact and is an acyclic resolution for $\mathcal{F}$. In particular, we have $\check{H}^i(\check{\mathcal{C}}^\bullet(\mathfrak{U},\mathcal{F}) = H^i(\check{C}^\bullet(\mathfrak{U},\mathcal{F}) = H^i(X,\mathcal{F})$. Now we want to build an acyclic resolution for the complex of sheaves $\mathcal{F}^\bullet$. We consider the double complex:

$$\begin{array}{ccccc}
 & \uparrow & & \uparrow & \\
\dots \longrightarrow & E^{p+1,q} = \check{\mathcal{C}}^{p+1}(\mathfrak{U},\mathcal{F}^q) & \xrightarrow{(-1)^{p+1}d^q} & E^{p+1,q+1} = \check{\mathcal{C}}^{p+1}(\mathfrak{U},\mathcal{F}^q) & \longrightarrow \dots \\
 & \uparrow{\scriptstyle \delta^p} & & \uparrow{\scriptstyle \delta^p} & \\
\dots \longrightarrow & E^{p,q} = \check{\mathcal{C}}^{p}(\mathfrak{U},\mathcal{F}^q) & \xrightarrow{(-1)^{p}d^q} & E^{p+1,q+1} = \check{\mathcal{C}}^{p}(\mathfrak{U},\mathcal{F}^{q+1}) & \longrightarrow \dots \\
 & \uparrow & & \uparrow & \\
 & \dots & & \dots &
\end{array}$$

Let $\mathrm{T}^\bullet$ be the total complex associated to the double complex above, that is

$$\mathrm{T}^n = \bigoplus_{p+q=n} E^{p,q} \xrightarrow[D]{} \mathrm{T}^{n+1} = \bigoplus_{p+q=n+1} E^{p,q}$$

with $(D^n(a))_{p,q} = \delta^{p-1,q}(a_{p-1,q}) + (-1)^p d^{p,q-1}(a_{p,q-1})$. We have that $F^\bullet \to \mathrm{T}^\bullet$ is an acyclic resolution and $\mathbb{H}^n(X,\mathcal{F}^\bullet) \cong H(\Gamma(X,\mathrm{T}^\bullet))$.

Example 6. *Let X/K be a smooth projective curve, $\Omega^1_{X/K}$ is locally free of rank one, and we have $\Omega^i_{X/K} := \bigwedge^i \Omega^1_{X/K} = 0$ for $i>1$. Then the de Rham complex is simply $\mathcal{O}_X \to \Omega^1_{X/K}$, and we have $H^i_{\mathrm{dR}}(X) \cong H^i(T^\bullet)$, where $T^\bullet$ is the total complex associated to the following Čech double complex: Let $\mathfrak{U} = \{U,V\}$ be an open cover by two affines (always possible for a projective curve).*

$$\begin{array}{ccc}
\check{\mathcal{C}}^0(\mathfrak{U},\Omega^1_{X/K}) = \Omega^1_{X/K}(U)\oplus\Omega^1_{X/K}(U) & \xrightarrow{\delta^0} & E^{p+1,q+1} = \check{\mathcal{C}}^1(\mathfrak{U},\Omega^1_{X/K}) = \Omega^1_{X/K}(U\cap V) \\
\uparrow{\scriptstyle d} & & \uparrow{\scriptstyle -d} \\
\check{\mathcal{C}}^0(\mathfrak{U},\mathcal{O}_X) = \mathcal{O}_X(U)\oplus\mathcal{O}_X(V) & \xrightarrow{\qquad\delta^0\qquad} & \check{\mathcal{C}}^p(\mathfrak{U},\mathcal{O}_X) = \mathcal{O}_X(U\cap V)
\end{array}$$

namely, we have

$$T^0 = \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \xrightarrow[D^0]{} T^1 = \Omega^1_X(U) \oplus \Omega^1_X(V) \oplus \mathcal{O}_X(U \cap V) \xrightarrow{D^1} T^2 = \Omega_X(U \cap V)$$

with $D^0(f_U, f_V) = (dF_U, df_V, f_U|_{U\cap V} - f_V|_{U\cap V})$ *and* $D^1(\omega_U, \omega_V, f_{UV}) = \omega_U|_{U\cap V} - \omega_V|_{U\cap V} - df_{UV}$. *Therefore, we have*

- $H^0_{\mathrm{dR}}(X) = \ker D^0 = \{(f_U, f_V) \in \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) | df_U = 0 = df_V \text{ and } f_U|_{U\cap V} = f_V|_{U\cap V}\} = \mathcal{O}_X(X) \cong K$
- $H^1_{\mathrm{dR}}(X) = \frac{\ker D^1}{\operatorname{Im} D^0} = \frac{(\omega_U,\omega_V,f_{UV}) \in \Omega^1_X(U)\oplus\Omega_X(V)\oplus\mathcal{O}_X(U\cap V) | \omega_U|_{U\cap V} - \omega_V|_{U\cap U} - df_{UV} = 0}{(df_U, df_V, f_U|_{U\cap V} - f_V|_{U\cap V} | (f_U, f_V) \in \mathcal{O}_X(U)\oplus\mathcal{O}_X(V))}$. *It has dimension* $2g$, *where* g *is the genus.*
- $H^2_{\mathrm{dR}}(X) = \frac{\Omega^1_X(U\cap V)}{\operatorname{Im} D^1}$

We can also define a relative version of de Rham cohomology:

Definition 11. *Let* $\pi : X \to S$ *be a smooth morphism of schemes over* K. *Then the relative de Rham cohomology is the sheaf* $H^i_{\mathrm{dR}}(X/S) = \mathbb{R}\pi_*\Omega^\bullet_{X/S}$ *on* S.

3.2 Connections

Connections will allow to have other coefficients for the de Rham cohomology that are not the structure sheaf. Let X be a smooth scheme over K, and $\mathcal{F} \in \mathrm{QCoh}(\mathcal{O}_X)$.

Definition 12. *A connection on* $\mathcal{F}$ *is a* K*-linear morphism* $\nabla : \mathcal{F} \to \Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{F}$ *satisfying on sections* $\nabla(af) = da \otimes f + a\nabla(f)$.

Observe that a connection gives rise to maps $\nabla_i : \Omega^i_{X/K} \otimes \mathcal{F} \to \Omega^{i+1}_X \otimes \mathcal{F}$ defined by $\omega \otimes f \mapsto d\omega \otimes f + (-1)^i \omega \wedge \nabla(f)$.

Definition 13. *Given* $(\mathcal{F}, \nabla)$ *a sheaf with connection, the curvature of* ∇ *is the* $\mathcal{O}_X$*-linear map* $K := \nabla_1 \circ \nabla : \mathcal{F} \to \Omega^2_X \otimes \mathcal{F}$. *One can check that, with this definition,* $()\nabla^{i+1} \circ \nabla^i)(\omega \otimes f) = \omega \wedge K(f)$.

Definition 14. *We say that the connection* ∇ *is integrable if* $K(\nabla) = 0$, *or, equivalently, if* ∇^i *give rise to a complex* $\mathcal{F} \xrightarrow[\nabla]{} \mathcal{F} \otimes \Omega^1_X \xrightarrow[\nabla^1]{} \mathcal{F} \otimes \Omega^2_X \to \dots$. *In that case, we denote this complex* $\Omega^\bullet_X(\mathcal{F})$, *and we define* $H^i_{\mathrm{dR}}(X, \mathcal{F}, \nabla) = \mathbb{H}^i(X, \Omega^\bullet_X(\mathcal{F}))$.

Remark 7. *A connection gives rise to a* K*-linear map* $\mathrm{Der}(\mathcal{O}_X) \to \mathrm{End}_K(\mathcal{F})$ *given by* $D \mapsto \nabla(D)$, *where* $\nabla(D) : \mathcal{F} \to \mathcal{F}$ *is the map* $\mathcal{F} \xrightarrow[\nabla]{} \mathcal{F} \otimes \Omega^1_X \xrightarrow[\varphi_D \otimes 1]{\mathcal{O}} \otimes \mathcal{F}$, *defined via the identification* $D \leftrightarrow \varphi_D$ *given by the isomorphism* $\mathrm{Der}_K(\mathcal{O}_X) \cong \mathrm{Hom}_K(\Omega^1_X, \mathcal{O}_X)$.

We should do this exercise for the Gauss-Manin connection.

Example 7.

1. $d : \mathcal{O}_X \to \Omega^1_X \cong \Omega^1_X \otimes \mathcal{O}_X$ *is a connection on* X.
2. *Let* $\mathcal{F} \cong \mathcal{O}_X$ *and let* $\omega \in H^0(X, \Omega^1_X)$. *Then* $\nabla(1) = \omega$ *defines a connection:* $\nabla(a) = \nabla(a1) = a\nabla(1) + da \otimes 1 \cong a\omega + da$.

3. *Let $\mathcal{F}$ be free of rank d over $\mathcal{O}_X$, i.e. $\mathcal{F} \cong \mathcal{O}_X^d$, and let $e_1, \dots, e_d \in H^0(X, \mathcal{F})$ be generators of $\mathcal{F}$. Let $\omega_1, \dots, \omega_d \in H^0(X, \Omega_X^1)$. Then, $\nabla(e_i) := \omega_i$ defines a connection on $\mathcal{F}$. Indeed, $\forall s \in \mathcal{F}(U) \cong \bigoplus_{i=1}^d \mathcal{O}_X(U) e_i$, $s = a_1 e_1|_U + \cdots + a_d e_d|_U$, $a_i \in \mathcal{O}_X(U)$. Then,*

$$\nabla(s) = \sum_{i=1}^d \nabla(a_i e_i|_U) = \sum_{i=1}^d da_i \otimes e_i|_U + a_i \omega_i \otimes e_i|_U$$

4. *Let $\mathcal{L}$ be a line bundle on X. Then, there exists a (finite, open) cover $\{U_i\}_{i \in I}$ that trivializes it: $\mathcal{L}|_{U_i} \cong \mathcal{O}_X|_{U_i} e_i$, for $e_i \in \mathcal{L}(U_i)$. Then, $e_i|_{U_i \cap U_j} = a_{ij} e_j|_{U_i \cap U_j}$, for some $a_{ij} \in \mathcal{O}_X^\times(U_{ij})$, and $a_{ik|_{U_{ijk}}} = a_{ij}|U_{ijk} a_{jk}|_{U_{ijk}}$. So, this defines a Čech cocycle, and we have a map*

$$\begin{aligned} \mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^\times) &\to H^1(X, \Omega_X^1) \\ \mathcal{L} \mapsto [(a_{ij})_{i,j}] &\mapsto \left[\left(\eta_{ij} = \frac{da_{ij}}{a_{ij}} \right)_{i,j} \right] \end{aligned}$$

Let $[\mathcal{L}] = 0$ on $H^1(X, \Omega_X^1)$, which implies $\exists (\omega_i)_{i \in I}$, with $\omega_i \in \Omega_X^1(U_i)$ such that $\frac{da_{ij}}{a_{ij}} = \eta_{ij} = \omega_i|_{U_{ij}} - \omega_j|_{U_{ij}}$. Then we set $\nabla_i(e_i) = \omega_i \otimes e_i$, and by the previous example, these define connections $\nabla_i : \mathcal{F}|_{U_i} \to \Omega_X^1|_{U_i} \otimes \mathcal{F}|_{U_i}$. We now check that the connections ∇_i glue and therefore define a connection on $\mathcal{F}$. Indeed, on intersections we have

$$\begin{aligned} &\nabla_i(e_i|_{U_{ij}}) = \omega_i \otimes e_i|_{U_{ij}} \\ &\nabla_j(e_i|_{U_{ij}}) = \nabla_j(a_{ij} e_j|_{U_{ij}}) = (a_{ij} \nabla_j(e_j) + da_{ij} \otimes e_j)|_{U_{ij}} = (a_{ij}(\omega_j \otimes e_j) + da_{ij} \otimes e_j)|_{U_{ij}} \end{aligned}$$

And now using $\omega_j|_{U_{ij}} = \omega_i|_{U_{ij}} - \eta_{ij}$, we get

$$\nabla_j(e_i|_{U_{ij}}) = (a_{ij}(\omega_i \otimes e_j) - a_{ij} \frac{da_{ij}}{a_{ij}} \otimes e_j)|_{U_{ij}} = \omega_i \otimes a_{ij} e_j|_{U_{ij}} = \omega_i \otimes e_i|_{U_{ij}} = \nabla_i(e_i|_{U_{ij}})$$

Remark: The cohomology class $[\mathcal{L}]$ is the obstruction to the existence of a connection.

3.3 Connections over $\mathbb{C}$

For this section we follow [6].

Definition 15. *Let X be a connected complex analytic variety. A complex local system on X is a sheaf of complex vector spaces that, locally on X, is isomorphic to the constant sheaf $\mathbb{C}^n$.*

Proposition 5. *Let V be a complex local system on X, and let $\mathcal{V} := \mathcal{O}_X \otimes_C V$. Then, there exists on the vector bundle $\mathcal{V}$ exactly one connection, called the canonical connection, whose horizontal sections (i.e. $\mathcal{V}^{\nabla=0}$) are the local sections of the subsheaf V. ∇ is integrable and*

$$\nabla(f \otimes_{\mathbb{C}} v) = df \otimes v \in \Omega_X^1 \otimes_{\mathbb{C}} V \cong \Omega_X^1 \otimes_{\mathbb{C}} \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{V} \cong \Omega_X^1 \otimes_{\mathcal{O}_X} \mathcal{V}$$

Proof. The $\mathbb{C}$-linear map ∇ clearly defines a connection: $\nabla(a(f \otimes v)) = \nabla(af \otimes v) = d(af) \otimes v = f da \otimes v + a df \otimes v = a\nabla(f \otimes v) + da(f \otimes v) \cong a\nabla(f \otimes v) + da \otimes (f \otimes v)$. Moreover $f \otimes v \in \mathcal{V}^{\nabla=0}(U)$ if and only if $\nabla(f \otimes v) = 0 \iff f \in \mathbb{C}$, that is, $\mathcal{V}^{\nabla=0} = \mathbb{C} \otimes_{\mathbb{C}} V \cong V$.

It is clearly integrable, as $(\nabla_1 \circ \nabla)(f \otimes v) = d^2 f \otimes v + df \otimes \nabla(1 \otimes v) = 0$, where we used that $d^2 f = 0$ and $1 \otimes v$ is an horizontal section. Finally, if ρ is another connection, it must satisfy $\rho(f \otimes v) = \rho(f \cdot 1 \otimes v) = df \otimes_{\mathcal{O}_X} (1 \otimes_{\mathbb{C}} v) + \rho(1 \otimes v) \cong df \otimes v$, where we used again that $1 \otimes v \in \mathcal{V}^{\nabla=0}$. □

This allows to prove the following theorem:

Theorem 3. *The following functors are inverse to one another and define an equivalence of categories:*

$$\left\{\begin{matrix}\textit{complex local systems}\\ V \textit{ on } X\end{matrix}\right\} \cong \left\{\begin{matrix}\textit{holomorphic vector bundles } \mathcal{V}\\ \textit{with integrable connection}\end{matrix}\right\}$$
$$V \longmapsto \mathcal{V} = (\mathcal{O}_X \otimes_{\mathbb{C}} V, d \otimes 1)$$
$$\mathcal{V}^{\nabla=0} \longmapsfrom (\mathcal{V}, \nabla)$$

Proof. (Sketch) The proposition above already shows part of the theorem. To conclude, let $\mathcal{V}$ be locally free of rank m, we need to show that $\mathcal{V}^{\nabla=0} \otimes_{\mathbb{C}} \mathcal{O}_X \cong \mathcal{V}$ and $\mathcal{V}^{\nabla=0}$ is a local system. Observe that locally, we can find $U \subset X$ isomorphic to an open subset of $\mathbb{C}^n$, where $z_1, \ldots, z_n$ are local coordinates, and $\mathcal{V}|_U \cong \mathcal{O}_X^m|U$. Let $e_1, \ldots, e_m \in \mathcal{O}_X(U)$ be a local basis. Then,

$$\nabla|_U : \mathcal{V}|_U \cong \mathcal{O}_X^m \to \mathcal{V}|_U \otimes \Omega^1_X|_U \cong \Omega_X|_U^m$$

Let $\Omega_U = (\omega_{ij})_{i,j}$ be a matrix of differential forms such that $\nabla|_U(e_i)$ is the ith column of Ω_U. Then for $y \in \mathcal{V}(U)$, $y = \sum_{i=1}^m y_i e_i$ and $\nabla(y) = \sum_{i=1}^m \nabla(y_i e_i) = \sum_{i=1}^m y_i \nabla(e_i) + dy_i \otimes e_i$. Then we have

$$\nabla(y) = \begin{pmatrix} dy_1 \\ \vdots \\ dy_m \end{pmatrix} + \Omega_U \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

Therefore we have that

$$y \in \mathcal{V}^{\nabla=0} \iff \begin{pmatrix} dy_1 \\ \vdots \\ dy_m \end{pmatrix} + \Omega_U \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = 0 \tag{1}$$

Now if we write in local coordinates $y_i = y_i(z_1, \ldots, z_n)$, $\omega_{ij} = \sum_{k=1}^n a_{ijk}(z_1, \ldots, z_n) dz_k$, 1 becomes

$$\frac{\partial}{\partial z_k} y_i(z_1, \ldots, z_n) + \sum_{j=1}^m a_{ijk}(z_1, \ldots, z_n) y_j(z_1, \ldots, z_n) = 0$$

and $\mathcal{V}^{\nabla=0}$ is described by the system of partial differential equations above. By integrability, they form a $\mathbb{C}$-vector space of dimension m. This shows that $\mathcal{V}^{\nabla=0}$ is a local system of dimension m. To conclude, observe that $\mathcal{V}^{\nabla=0} \otimes_{\mathbb{C}} \mathcal{O}_X \cong \mathcal{V}$ since they're locally free $\mathcal{O}_X$-modules of the same rank contained one in the other. □

Remark 8. *If V is a complex local system on X, $\mathcal{V} = V \otimes_{\mathbb{C}} \mathcal{O}_X$, then $\Omega^\bullet_X \otimes_{\mathbb{C}} V \cong \Omega^\bullet_X \otimes_{\mathcal{O}_X} \mathcal{O}_X \otimes_{\mathbb{C}} V \cong \Omega^\bullet_X \otimes_{\mathcal{O}_X} \mathcal{V}$. Thus we have $\mathbb{H}^n(X, \Omega^\bullet_X \otimes_{\mathbb{C}} V) \cong \mathbb{H}^n(X, \Omega^\bullet_X \otimes_{\mathcal{O}_X} \mathcal{V})$. By Poincaré Lemma, $0 \to \underline{\mathbb{C}} \to \Omega^1_X \to \Omega^2_X \to \ldots$ is exact, so $0 \to V \to \Omega^\bullet_X \otimes_{\mathbb{C}} V$ is an acyclic resolution of V, and therefore we have*

$$H^1_{\mathrm{dR}}(X, \mathcal{V}, \nabla) = \mathbb{H}^i(X, \Omega^\bullet_X \otimes_{\mathbb{C}} V) \cong H^i(X, V)$$

in other words: The sheaf cohomology of V computes the de Rham cohomology of $\mathcal{V}$.

3.4 Gauss-Manin connection

The reference for this section is [8]. Let $\pi : X \to S$ be a smooth morphism of schemes over K. Then we can define a connection

$$\nabla : H^q_{\mathrm{dR}}(X/S) \to \Omega^1_S \otimes_{\mathcal{O}_S} \otimes_{\mathcal{O}_S} H^q_{\mathrm{dR}}(X/S)$$

which is called the Gauss-Manin connection. To construct it, we need some facts about homological algebra.

Let $\mathcal{A}$, be an abelian category, and $(C^\bullet, d^\bullet)$ be a complex with bounded filtration $F^\bullet C^\bullet$, that is, for each q, we have

$$0 \subseteq \cdots \subseteq F^p C^q \subseteq F^{p-1} C^q \subseteq \cdots \subseteq F^0 C^q = C^q$$

with differentials preserving the filtration, i.e. $d^q : F^p C^q \to F^p C^{q+1}$. Then the differentials give a complex $\mathrm{gr}^\bullet C^\bullet$ of graded pieces $(F^p C^\bullet, d^\bullet)$, $\mathrm{gr}^p C^q = \frac{F^p C^q}{F^{p+1} C^q}$. Moreover, we have a filtration on cohomology:

$$F^p H^q(C^\bullet) = \mathrm{Im}(H^q(F^p C^\bullet) \to H^q(C^\bullet))$$

Lemma 4. *The filtration $F^p C^\bullet$ determines a spectral sequence with $E_0^{p,q} = \mathrm{gr}^p C^{p+q}$, $E_1^{p,q} = H^{p+q}(\mathrm{gr}^p(C^\bullet))$ and $E_\infty^{p,q} = H^{p+q}(C^\bullet)$, and page 1 maps $d_1^{p,q} : H^{p+q}(\mathrm{gr}^p(C^\bullet)) \to H^{p+q+1}(\mathrm{gr}^{p+1}(C^\bullet))$, forresponding to the connecting map on cohomology arising from the short exact sequence*

$$0 \to \mathrm{gr}^{p+1} C^\bullet \to F^p C^\bullet / F^{p+1} C^\bullet \to \mathrm{gr}^p C^\bullet \to 0$$

Lemma 5. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $\mathcal{A}$ with enough injectives, and $Ain\mathcal{A}$ an object with filtration $F^\bullet A$. Let $T : \mathcal{A} \to \mathcal{B}$ be a left exact functor. We have a spectral sequence*

$$E_1^{p,q} = R^{p+q} T(\mathrm{gr} A) \implies R^{p+q} T(A)$$

and page 1 maps arising from $0 \to \mathrm{gr}^{p+1} C^\bullet \to F^p C^\bullet / F^{p+1} C^\bullet \to \mathrm{gr}^p C^\bullet \to 0$.

Now we will apply Lemma 5 to our situation. Under our assumptions, we have a split exact sequence of $\mathcal{O}_X$-modules

$$0 \to \pi^* \Omega^\bullet_S \to \Omega^\bullet_X \to \Omega^\bullet_{X/S} \to 0$$

we regard $\pi^* \Omega^\bullet_S \hookrightarrow \Omega^\bullet_X$ and define a filtration, called the Koszul filtration

$$F^p \Omega^\bullet_X = \mathrm{Im}(\pi^* \Omega^p_S \otimes \Omega^{\bullet - p}_X \to \Omega^\bullet_X)$$

which has graded pieces $\mathrm{gr}^p \Omega^\bullet_X = \pi^* \Omega^p_S \otimes \Omega^{\bullet - p}_{X/S}$. Applying Lemma 4, we get a spectral sequence

$$E_1^{p,q} = \mathbb{R}^{p+q} \pi_* (\mathrm{gr}^p \Omega^\bullet_X) \implies \mathbb{R}^{p+q} \pi_* \Omega^\bullet_X$$

and observe that we have $E_1^{p,q} = \mathbb{R}^{p+q} \pi_* (\mathrm{gr}^p \Omega^\bullet_X) = \mathbb{R}^{p+q} \pi_* (\pi^* \Omega^p_S \otimes_{s\mathcal{O}_X} \Omega^{\bullet - p}_{X/S}) = \Omega^p_S \otimes \mathbb{R}^{p+q} \pi_* \Omega^{\bullet - p}_{X/S} = \Omega^p_S \otimes \mathbb{R}^q \pi_* \Omega^\bullet_{X/S} = \Omega^p_S \otimes H^q_{\mathrm{dR}}(X/S)$. and the page one maps are given by $E_1^{p,q} = H^q_{\mathrm{dR}}(X/S) \xrightarrow[d_1^{0q}]{} \Omega^1_S \otimes H^q_{\mathrm{dR}}(X/S) \xrightarrow[d_1^{1q}]{} \Omega^2_S \otimes H^q_{\mathrm{dR}}(X/S) \to \ldots$.

We let $\nabla = d_1^{0,q}$. If ∇ is a connection, we get integrability for free, from the multiplicative structure of $\{E_r^{pq}\}$: It is a general fact that, if $\{E_r^{p,q}\}_{r\geq a}$ is a spectral sequence and at page $r = a$ we have a bigraded product $E_a^{p,q} \times E_a^{p',q'} \to E_a^{p+p',q+q'}$ such that the differentials satisfy

$$d_a^{p+p',q+q'}(x \cdot x') = d_a^{p,q}(x)x' + (-1)^p d_r^{p',q'}(x')x \tag{2}$$

then the product of two cocycles/coboundaries is still a cocycle/coboundaty, and the same relations extend to cohomology, and so to page r for $r \geq a$.

In our case, we have $\Omega_X^p \otimes \Omega_X^q \to \Omega_X^{p+q}$ satisfying $d(\omega \wedge \tau) = d\omega \wedge \tau + (-1)^p \omega \wedge \tau$ and it preserves filtrations, so it gives a bigraded product on the spectral sequence, and page 1 maps will satisfy a relation like 2. Moreover, we have $E_1^{p,0} = \Omega_S^p \otimes \mathcal{O}_S$, so we can see the de Rham complex $(\Omega_S^\bullet, d^\bullet)$ as a subcomplex of $(E_1^{\bullet,0}, d_1^{\bullet,0})$. The multiplicative structure in the spectral sequence gives $E_1^{0,0} \times E_1^{0,q} \to E_1^{0,q}$ with $E_1^{0,q} = H_{\mathrm{dR}}^q(X/S)$ and $E_1^{0,0} = \mathcal{O}_S$, with $d_1^{0,q}(af) = d_1^{0,0}(a)f + ad_1^{0,q}(f)$, or, in other words, as we have defined $\nabla = d_1^{0,q}$, $\nabla(af) = da \otimes f + a\nabla(f)$, and this shows that ∇ is indeed a connection on $H_{\mathrm{dR}}^q(X/S)$.

4 Infinitesimal site

Speaker: Anna, 09/10/26

We will define the infinitesimal site and show that it is indeed a site. This section follows [2] (with the difference that we work over schemes instead of rigid spaces).

Definition 16. *A site is the datum $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$, with $\mathcal{C}$ a category with fibred products and $\mathrm{Cov}(\mathcal{C})$ a family of morphisms in $\mathcal{C}$ for every object $U \in \mathcal{C}$: $\{U_i \to U\}_{i\in I}$, satisfying*

i) $\{U \xrightarrow[\mathrm{id}]{} U\} \in \mathrm{Cov}(\mathcal{C})$.

ii) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})$ and for every $i \in I$, $\{U_{ij} \to U_i\}_{j\in J_i} \in \mathrm{Cov}(\mathcal{C})$, then $\{U_{ij} \to U\}_{i\in I, j\in J_i} \in \mathrm{Cov}(\mathcal{C})$

iii) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})$, and $V \to U$ is any morphism, $\{U_i \times_U V \to V\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})$.

Example 8. *1. X topological space, $\mathcal{C} = \mathrm{Open}(X)$ and $\mathrm{Cov}(\mathcal{C})$ are the usual coverings, $\{U_i\}_{i\in I}$ such that $\bigcup U_i 0 U$.*

2. X scheme, the Zariski site on X has as objects $U \to X$ open immersions.

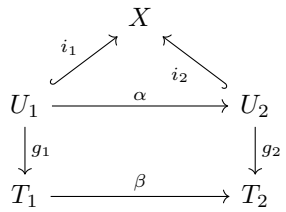
Now let K be a field of characteristic 0, and X a smooth scheme over K.

Definition 17. *Let $g : U \to T$ morphism of K-schemes. We say that g is a nilpotent closed immersion if it is a closed immersion defined by an ideal $\mathcal{I} \subseteq \mathcal{O}_T$ that is locally nilpotent, that is, for every $t \in T$, there exists $U \ni t$, $U \subseteq T$ open such that $\mathcal{I}|_U^n = 0$.*

Definition 18. *The infinitesimal category $(X/K)_{\mathrm{inf}}$ is the following:*

- *Objects: Pairs (U, T) with U, T K-schemes, $U \hookrightarrow X$ open immersion and $g : U \to T$ nilpotent closed immersion.*

- *Morphisms:* *Pairs $(\alpha, \beta) : (U_1, T_1) \to (U_2, T_2)$ such that $\alpha : U_1 \to U_2$ is an open immersion of K-schemes, and β is such that the following diagram commutes:*



Example 9. *1.* $(X, X) \in (X/K)_{\text{inf}}$.

2. *The diagonal map $U = \operatorname{Spec} B \subseteq U \times_K U$ closed immersion, is induced by the morphism $\mu : B \otimes B \to B$ given by $b \otimes b' \mapsto bb'$. We have $\ker \mu = I = \langle b_i \otimes 1 - 1 \otimes b_i \rangle$. Take $\operatorname{Spec} B = U \hookrightarrow \operatorname{Spec}(B \otimes B/I^{m+1}) = \mathcal{P}^m$. Then the pair $(U, \mathcal{P}^m_U) \in (X/K)_{\text{inf}}$.*

3. *More generally, for U not necessarily affine, take $\mathcal{P}^m = (|U|, \mathcal{O}_U \otimes \cdots \otimes \mathcal{O}_U/\mathcal{I}^{m+1})$.*

4. *Observe that $(X, \mathcal{P}^m) \rightrightarrows (X, X)$ given by (locally for affines) $A \rightrightarrows A \otimes A$, $a \mapsto a \otimes 1$ and $a \mapsto 1 \otimes a$ give two different maps to (X, X), which would be the natural final object to consider. This shows an example of the fact mentioned in the first section: The infinitesimal site does not have a terminal object.*

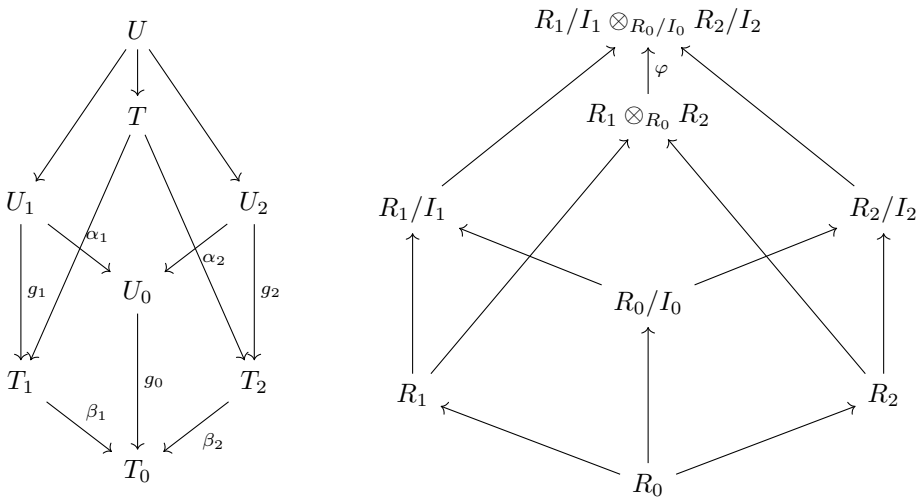
Proposition 6. *$(X/K)_{\text{inf}}$ has fiber products: If $(\alpha_1, \beta_1) : (U_1, T_1) \to (U_0, T_0)$ and $(\alpha_2, \beta_2) : (U_2, T_2) \to (U_0, T_0)$, the fiber product is given by $(U, T) := (U_1 \times_{U_0} U_2, T_1 \times_{T_0} T_2)$.*

Proof. We first show that the object $(U, T) \in (X/K)_{\text{inf}}$. For that we will use the following lemma:

Lemma 6. *Let R_0 be a commutative ring, M_1, M_2 be R_0-modules, and $I_0 \subseteq R_0$, $I_i \subseteq M_i$ R_0-submodules, such that $I_0 M_i \subseteq I_i$ (for $i = 1, 2$). Then,* $\ker(M_1 \otimes_{R_0} M_2 \to M_1/I_1 \otimes M_2 \to M_1/I_1 \otimes M_2/I/2) = I_1 \otimes M_2 + M_1 \otimes I_2$.

Now let's proceed to the proof of the proposition. First observe that, as $U_2 \to U_0$ is an open immersion, the base change $U_1 \times_{U_0} U_2 \to U_1$ is also an open immersion, and therefore also the

composition $U_1 \times_{U_0} U_2 \to U_0$ is. Now consider the following diagrams:



where the second diagram is given by considering $\operatorname{Spec} R_i \subseteq T_i$ and $g_i|_{\operatorname{Spec} R_i}$. We have $I_i \subseteq R_i$ satisfying $I_i^n = 0$. Observe that the condition in the lemma $I_0 M_i \subseteq I_i$ is imposed so that $\otimes_{R_0}$ and $\otimes_{R_0/I_0}$ are isomorphic. By the lemma, $\varphi : R_1 \otimes_{R_0} R_2 \to R_1/I_1 \otimes_{R_0/I_0} R_2/I_2$ is surjective and has nilpotent kernel, and therefore we have that $(U, T) \in (X/K)_{\text{inf}}$.

Now we have to check that (U, T) satisfies the universal property of the fibred product. But U and T respectively satisfy it, as they're defined as fibred products of schemes, so if (U', T') is such that we have maps $(U', T') \to (U_i, T_i)$ such that they agree when composed with $(U_i, T_i) \to (U_0, T_0)$, we get unique maps $U' \to U$ and $T' \to T$, and the only thing that remains to check is that the diagram

$$\begin{array}{ccc} U' & \longrightarrow & U \\ \downarrow & & \downarrow \\ T' & \longrightarrow & T \end{array}$$

commutes. But this follows from the fibre product property of T: Indeed, the maps $U' \to T' \rightrightarrows T_i$ agree when composed with $T_i \to T_0$, so there is a unique factorization of it through T, and therefore $U' \to U \to T$ and $U' \to T' \to T$ must be the same morphism. □

Now we deal with coverings in $(X/K)_{\text{inf}}$

Definition 19. *A covering in $(X/K)_{\text{inf}}$, $\{(\alpha_i, \beta_i) : (U_i, T_i) \to (U, T)\}_{i \in I}$ such that $\{\alpha_i : U_i \to U\}_{i \in I}$ and $\{\beta_i : T_i \to T\}_{i \in I}$ are respectively coverings for the Zariski topologies for U and T.*

It is immediate to see that they satisfy the axioms of coverings on a site, it follows from those of the Zariski sites for U and T.

Definition 20. *Let $\mathcal{C}$ be a category, and $A, B \in \operatorname{Ob}(\mathcal{C})$, $f, g : A \rightrightarrows B$ morphisms. The equalizer of f and g is a pair (E, e), $e : E \to A$ satisfying that $fe = ge$ and it is universal with respect to*

that property: $\forall D \in \mathrm{Ob}(\mathcal{C})$ *and* $\varphi : D \to A$ *such that* $f\varphi = g\varphi$*, there exists* $D \to E$ *making the following diagram commute:*

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ \uparrow & \nearrow_{\varphi} & \\ D & & \end{array}$$

Proposition 7. $(X/K)_{\mathrm{inf}}$ *has equalizers. In particular, if* $(\alpha_i, \beta_i) : (U_1, T_1) \rightrightarrows (U_2, T_2)$*, for* $i = 1, 2$*, the equalizer is given by* $(U.T) = (Eq(\alpha_1, \alpha_2), Eq(\beta_1, \beta_2))$*.*

Proof. First remark that if we have $f, g : X \rightrightarrows Y$ morphism of K-schemes, in the category of K-schemes their equalizer is given by the fibre product of the diagrams

$$\begin{array}{ccc} \mathrm{Eq}(f,g) & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\Delta_Y} & Y \times_K Y \end{array}$$

Where the map $X \to Y \times_K Y$ is given by (f, g). Indeed, take $Z \underset{\varphi}{\to} X \rightrightarrows Y$ with $f\varphi = g\varphi$, we have maps $\varphi : Z \to X$ and $f\varphi = g\varphi : Z \to Y$, and we get a unique map $Z \to Eq(f, g)$ by the property of the fibred product. So the object (U, T) exists as defined in the statement, and we just have to check that there exists $g : U \to T$ nilpotent closed immersion. First, the existence of g is clear by the equalizer property of T.

Now we have to see that g is a nilpotent closed immersion: For that, we use again Lemma 6: Consider the diagrams that define the equalizers:

$$\begin{array}{ccc} U & \longrightarrow & U_1 \\ \downarrow & & \downarrow \\ U_2 & \longrightarrow & U_2 \times_K U_2 \end{array} \qquad \begin{array}{ccc} T & \longrightarrow & T_1 \\ \downarrow & & \downarrow \\ T_2 & \longrightarrow & T_2 \times_K T_2 \end{array}$$

these correspond locally to diagrams

$$\begin{array}{ccc} R_1/I_1 \otimes_{R_2/I_2 \otimes R_2/I_2} R_2/I_2 & \longleftarrow & R_1/I_1 \\ \uparrow & & \uparrow \\ R_2/I_2 & \longleftarrow & R_2/I_2 \times_K R_2/I_2 \end{array} \qquad \begin{array}{ccc} R_1 \otimes_{R_2 \otimes R_2} R_2 & \longleftarrow & R_1 \\ \uparrow & & \uparrow \\ R_2 & \longleftarrow & R_2 \times_K R_2 \end{array}$$

and g corresponds on affines to the map $R_1 \otimes_{R_2 \otimes R_2} R_2 \to R_1/I_1 \otimes_{R_2/I_2 \otimes R_2/I_2} R_2/I_2$, which is surjective, and nilpotent by Lemma 6. □

Remark 9. *Observe that the proof of the existence of equalizers as shown for* K*-schemes can be used to prove that equalizers exist in any category where we have fibred products and products. The latter is not true for* $(X/K)_{\mathrm{inf}}$ *(recall that there is no terminal object), which is why we were forced to use a slightly more complicated argument.*

5 Presheaves and sheaves on the infinitesimal site

Speaker: Michael, 23/10/26 and 30/10/26.

5.1 Sheaves on a site

In what follows, let $\mathcal{C}$ be a site.

Definition 21. *The category of presheaves on $\mathcal{C}$ is the functor category $\mathcal{P}(\mathcal{C}) := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$. We say that a presheaf $F \in \mathcal{P}(\mathcal{C})$ is a sheaf if, for all cover $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})$, we have that the following diagram is exact*

$$F(U) \longrightarrow \prod_{i\in I} F(U_i) \rightrightarrows \prod_{(i,j)\in I^2} F(U_i \times_U U_j)$$

we denote by $\mathcal{S}(\mathcal{C})$ the full subcategory of sheaves on $\mathcal{C}$.

One can also define presheaves with values in other categories, usually Ab. In fact, a (pre)sheaf with values in abelian groups is nothing but a group object in the category (pre)sheaves with values in Set. This follows from the following fact:

Lemma 7. *$\mathcal{P}(\mathcal{C})$ is complete and cocomplete (that is, all limits and colimits exist), and $\mathcal{S}(\mathcal{C})$ is complete, and limits are computed in $P(\mathcal{C})$.*

Proof. The category Set has all limits and colimits, therefore the same is true for $\mathcal{P}(\mathcal{C})$, inherited from Set pointwise. $\mathcal{S}(\mathcal{C})$ is complete because the condition of being a sheaf is defined by an equalizer (a limit), and limits commute. □

Recall Grothendieck's AB-properties: Let $\mathcal{A}$ be an abelian category.

- AB3: $\mathcal{A}$ has direct sums.
- AB4: $\mathcal{A}$ satisfies AB3 and direct sums are exact.
- AB5: $\mathcal{A}$ satisfies AB3 and filtered colimits are exact.

similarly we have the dual notions:

- AB3*: $\mathcal{A}$ has products.
- AB4*: $\mathcal{A}$ satisfies AB3* and products are exact.
- AB5*: $\mathcal{A}$ satisfies AB3* and cofiltered limits are exact.

Corollary 1. *$\mathcal{P}_{\mathrm{ab}}(\mathcal{C}) := \mathrm{Ab}(\mathcal{P}(\mathcal{C})) \cong \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Ab})$ is complete and cocomplete, is abelian and satisfies AB5, AB4*. $\mathcal{S}_{\mathrm{ab}}(\mathcal{C}) := \mathrm{Ab}(S(\mathcal{C}))$ is complete and additive.*

Observe that we don't obtain cocompleteness of $\mathcal{S}$ immediately. This will be a consequence of the sheaffification formalism that we explain in what follows.

Theorem 4. *([1], Chapter II, Theorem 1.1) The inclusion $i : \mathcal{S}(\mathcal{C}) \to \mathcal{P}(\mathcal{C})$ admits a left adjoint functor $(-)^{\#}$, called the sheaffification functor. That is, $(-)^{\#} : \mathcal{P}(\mathcal{C}) \to \mathcal{S}(\mathcal{C})$ satisfying $\mathrm{Hom}_{\mathcal{S}(\mathcal{C})}(P^{\#}, F) \cong \mathrm{Hom}_{\mathcal{P}(\mathcal{C})}(P, iF)$, for every presheaf P and sheaf F. The sheaffification functor is exact.*

Proof. (Sketch) This is Theorem 1.1For every $U \in \mathcal{C}$, let J_U be the category where objects are coverings $\{U_i \to U\}_{i\in I}$ and morphisms are given by refinement. Let $P \in \mathcal{P}(\mathcal{C})$. We have a functor $P_U : J_u^{\mathrm{op}} \to \mathrm{Set}$ defined by

$$P_U(\{U_i \to U\}) = \mathrm{Eq}\left(\prod_j P(U_i) \to \prod_{i,j} P(U_i \times_U U_j)\right)$$

Observe that this is would correspond to $\check{H}^0$ if P is valued in Ab (see for example how sheaffification is defined for rigid spaces, [4], 5.2). Now we define $P_U^+ := \mathrm{colim}_{J_U} P_U = \mathrm{colim}_{\{U_i\to U\}} P_U(\{U_i \to U\})$, if such a colimit exists. This factors through the category $J_U/\{\text{equivalent coverings}\}$, where two coverings are defined to be equivalent if they refine each other. $J_U/\{\text{equivalent coverings}\}$ is a filtered category. Then $(-)^+$ is left exact because it is defined as a limit (left exact, recall in general the definition of left exact in a non-abelian category is commuting with finite limits), followed by a filtered colimit (which are also left exact). Now we have:

Surprise ([1], II.1.2): $(P^+)^+$ is a sheaf.

This is shown in the following way: We first define a property:

Definition 22. *We say that a presheaf P satisfies property (+) if for every $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathcal{C})$, we have $P(U) \to \prod_i P(U_i)$ injective.*

We have the following properties of (+):

Lemma 8. *Let P be a presheaf. Then,*

1. *P^+ satisfies (+).*
2. *If P satisfies (+), then P^+ is a sheaf.*

Then, we define $P^\# := (P^+)^+$. Observe that for every covering $\{U_i \to U\}$ there is a natural map $P(U) \to P_U(\{U_i \to U\})$ given by $P(U) \to \prod_i P(U_i)$, and this induces a morphism of functors $P \to P^+$. Therefore this allows us to get a morphism $P \to P^\#$. As the sheaf condition is precisely the same as the definition of P^+, this tells us that $F \to F^\#$ is an isomorphism if F was already a sheaf. Observe that $P \to P^+$ (and therefore also $P \to P^\#$) is actually functorial. Then, this also gives the universal property of sheafification: Any map $P \to F$, for F a sheaf, must factor through $P^\#$. □

Remark 10. *For schemes (or adic spaces) sheaffification is defined using stalks, but in this more general context we don't necessarily have stalks, that's why we need to use this formalism.*

Corollary 2. *$\mathcal{S}(\mathcal{C})$ is cocomplete, and colimits are computed in $\mathcal{P}(\mathcal{C})$ and then sheafifying the result. $\mathcal{S}_{\mathrm{ab}}(\mathcal{C})$ is an abelian category and satisfies AB5, AB3*.*

5.2 Morphisms of sites

Definition 23. *Let $\mathcal{C}, \mathcal{D}$ be sites. A morphism $f : \mathcal{C} \to \mathcal{D}$ is a functor $\tilde{f} : \mathrm{Cat}(\mathcal{D}) \to \mathrm{Cat}(\mathcal{C})$ (remark that it goes in the reverse direction!) satisfying:*

- *$\tilde{f}$ sends covers to covers.*
- *$\tilde{f}$ preserves all (relevant) fibre products.*

Example 10. *If $f : X \to Y$ is a continuous map of topological spaces, then the functor $\tilde{f} : \mathrm{Open}(Y) \to \mathrm{Open}(X)$ given by $U \mapsto f^{-1}(U)$ defines a morphism of sites $\mathrm{Open}(X) \to \mathrm{Open}(Y)$.*

Proposition 8. *Let $f : \mathcal{C} \to \mathcal{D}$ be a morphism of sites. Then the functor $f_P : \mathcal{P}(\mathcal{C}) \to \mathcal{P}(\mathcal{D})$ given by $- \circ f$ (that is, $f_p(P)(X) \in \mathcal{P}(\mathcal{D})$ is given by $X \mapsto P(f(X))$) is exact. It restricts to a functor $f_* : \mathcal{S}(\mathcal{C}) \to \mathcal{S}(\mathcal{D})$ and admits a left adjoint given by $f^p : \mathcal{P}(\mathcal{D}) \to \mathcal{P}(\mathcal{C})$ given by $P \mapsto \left(U \mapsto \mathrm{colim}_{U \to \tilde{f}(V)} P(V)\right)$. This is called the Kan extension.*

Proof. We simply observe that the restriction f_* is well defined (i.e. it lands in $\mathcal{S}(\mathcal{D})$) precisely because we have required that the morphism of sites f preserves covers and fiber products. □

Remark 11. *If $\mathcal{C}$ has finite limits, then the colimit in the definition of the Kan extension is filtered, and therefore f^p is exact (left exact by this property and is always right exact by the adjunction).*

Corollary 3. *Let $f : \mathcal{C} \to \mathcal{D}$ be a morphism of sites. Then f_* admits a left adjoint $f^{-1}(-) := (f^p(-))^\#$. It is exact if f^p is.*

Example 11. *If X is a topological space, and $f : U \hookrightarrow X$ open, in addition to the morphism of sites as in 10:*

$$\begin{aligned} f : \mathrm{Open}(U) &\to \mathrm{Open}(X) \\ U \cap V &\longleftarrow\!\shortmid V \end{aligned}$$

we also have another morphism

$$\begin{aligned} u : \mathrm{Open}(X) &\to \mathrm{Open}(U) \\ f(V) &\longleftarrow\!\shortmid V \end{aligned}$$

It is easy to see that $f^p \cong u_p$, which is tipically denoted $(-)|_U$, the restriction to the open subspace U. This shows that f^p preserves sheaves in this special case, and it also admits an exact left adjoint, u^p, which is the extension by zero, and usually denoted $f_!$.

References

[1] M. Artin. *Grothendieck Topologies: Notes on a Seminar*. Harvard University, Department of Mathematics, 1962.

[2] Marco Baracchini. *BGG Decomposition for de Rham Sheaves on the Modular Elliptic Curve*. PhD thesis, Università degli Studi di Padova, 2024.

[3] Pierre Berthelot and Arthur Ogus. *Notes on Crystalline Cohomology. (MN-21)*. Princeton University Press, 1978.

[4] Siegfried Bosch. *Lectures on Formal and Rigid Geometry*. Lecture Notes in Mathematics. Springer, 2014.

[5] Marco d'Addezio. Differential operators. Notes from a talk avaliable online at `https://daddezio.pages.math.cnrs.fr/files/differential-operators.pdf`.

[6] Pierre Deligne. *Equations Différentielles à Points Singuliers Réguliers*. Lecture Notes in Mathematics. Springer, Berlin, Germany, 1970 edition, December 1970.

[7] Haoyang Guo. Crystalline cohomology of rigid analytic spaces. *Bull.Soc.Math.France*, 153 (4):831–948, 2025.

[8] Nicholas M. Katz and Tadao Oda. On the differentiation of De Rham cohomology classes with respect to parameters. *Journal of Mathematics of Kyoto University*, 8(2):199 – 213, 1968.